

Magnify

The documentation required by clause 7.10 of the Technical Specification of CIS for VSDC. It states every function the system performs, so that the system performs no function beyond those stated.

PRODUCT Magnify	VERSION 1.1.50	VENDOR YoFinder Ltd	TIN 149047989
SDC ROUTE Local VSDC	DOCUMENT Version 1		

Sections 1 to 5 and 7 to 10 describe the software as built. Section 6, the connection to the Sales Data Controller, is specified and built to the API specification but cannot be completed or verified without a controller instance; that section states precisely what is implemented and what is not. This document is revised for every change to the software, as clause 7.10 requires.

1 · PURPOSE AND SCOPE

Magnify is a point-of-sale and inventory management system for restaurants. It records orders, settles bills, prints receipts, tracks stock and produces management and fiscal reports.

This document covers Magnify 1.1.50, the fiscal build, which operates as a Certified Invoicing System connected to a Sales Data Controller. Non-fiscal builds of Magnify exist for use outside Rwanda; they are a different product and are not covered here.

Intended users

ROLE	USES	CAN
Waiter	Tablet or till	Take orders, send to kitchen, present bills
Supervisor	Till	All of the above, plus approve cancellations, discounts, refunds and cash withdrawals
Manager	Manager application	Menu, stock, staff, reports, fiscal configuration
Owner	Manager application	All of the above across every station
Auditor	Audit interface	Read software settings, records and counter state

2 · SYSTEM ARCHITECTURE

Magnify is three applications sharing one database. Only one of them issues receipts.

APPLICATION	PLATFORM	ROLE IN THE CIS
Till Magnify POS	Windows · Electron	This is the Certified Invoicing System. It settles bills, calls the VSDC and prints receipts. It is the only component that issues a receipt.
Manager	Web · Electron	Configuration, menu, stock, staff and reports. Does not issue receipts.
Tablet	Android	Order entry only. Has no printer and issues no receipt. Orders are sent to a till, which settles them.

The certified surface is the till alone. The tablet cannot print and therefore cannot issue a receipt; the manager application does not settle bills. Confining receipt issuance to one component keeps the certified boundary unambiguous.

Data flow for a sale

```
Waiter enters order → Kitchen ticket printed
                    → Bill presented (proforma)
                    → Till settles the bill
                      → journal entry written (PENDING)
                      → sale sent to local VSDC
                      → VSDC returns SDC id, receipt numbers,
                        internal data and signature
                      → journal entry confirmed (SENT)
                      → fiscal receipt printed
                    → VSDC relays to the RRA EBM server
```

The receipt is printed only after the VSDC has answered (clause 10). Should the VSDC not answer, no receipt is printed and the sale is not completed.

3 · INSTALLATION AND FISCAL CONFIGURATION

The fiscal build is installed from a signed installer under its own application identity, separate from non-fiscal builds of Magnify. The two cannot update into one another.

Required configuration

Until every field below is set, the till refuses to trade. It does not fall back to trading without VAT.

SETTING	LEVEL	PURPOSE
TIN	Taxpayer	Printed on every receipt
MRC	Outlet	Machine Registration Code (clause 7.4)
SDC ID	Outlet	Identifies the controller on the receipt
Branch code	Outlet	Outlet identifier issued by RRA
VSDC address	Outlet	Where the till reaches the controller

Changing the taxpayer

The TIN cannot be edited. It may be changed only in service mode, and only as part of a reset which erases every fiscal receipt, all receipt numbering and the cash drawer history held under the previous TIN (clause 7.2). Receipt numbering then recommences from 1 (clause 7.3).

The operation is refused unless four conditions are met: the system is in service mode, the fiscal records have been **exported and delivered to the taxpayer**, a named supervisor approves, and the erasure is explicitly confirmed. Re-entering the TIN already in use is refused, so that saving an unchanged settings form cannot trigger a reset.

The export is what reconciles this reset with the retention rule. Fiscal records are not removable by any other function. This one is required by clause 7.2 for a transfer of ownership, and it destroys records the taxpayer remains obliged to produce — so nothing is erased that has not first been placed in the taxpayer's own hands.

4 · FUNCTIONS

Clause 7.10 requires that the system have no functions beyond those stated here. This section is therefore exhaustive, including functions unrelated to invoicing.

4.1 Sales and service

FUNCTION	WHERE	NOTES
Take an order	Till, tablet	Against a table or as takeaway
Send to kitchen or bar	Till, tablet	Prints a numbered preparation slip per station
Amend an unsettled order	Till, tablet	Add, remove or move lines before settlement
Move a line between bills	Till	Supervisor approval; reversible
Remove a sent dish	Till	Supervisor approval; records no waste
Join bills	Till	Supervisor approval when waiters differ
Apply a line discount	Till	Supervisor approval; 0–100 per cent

FUNCTION	WHERE	NOTES
Cancel an unsettled order	Till, tablet	Supervisor approval and a reason
Settle a bill	Till	Issues the fiscal receipt
Refund a settled bill	Till	Supervisor approval; once per bill
Reprint a receipt	Till	Prints as a marked COPY
Open and close a service shift	Till	Supervisor; sets the business date

4.2 Cash

FUNCTION	WHERE	NOTES
Record an opening float	Till	Reported as the opening deposit
Record a cash deposit	Till	Clause 7.12
Record a cash withdrawal	Till	Supervisor approval required

4.3 Menu and stock

FUNCTION	WHERE	NOTES
Maintain dishes and prices	Manager	Item code, classification, tax type, unit
Classify an item as goods or service	Manager	Governs the stock check (clause 7.30)
Maintain recipes	Manager	Links a dish to its ingredients
Receive stock	Manager	Purchases, with FIFO batches
Record waste	Manager	With reason
Stock take	Manager	Counted against expected
Mise en place	Manager, till	Batch preparation consumed before ingredients
Closing stock at a date	Manager	Clause 7.31

4.4 Reporting

FUNCTION	WHERE	NOTES
Z daily report	Till, manager	Closes the day (clause 7.6)
X daily report	Till, manager	Since the last Z
Sales and item reports	Manager	Management information
Transactions and accounts	Manager	Journal entries, receivable and payable
Audit interface	Manager	Clause 7.26

4.5 Administration

FUNCTION	WHERE	NOTES
Staff and roles	Manager	Including supervisor approval PINs
Tables and stations	Manager	Floor plan
Printer configuration	Till, manager	Per device
Fiscal configuration	Manager	Section 3
Backup and restore	Manager	Excludes images and chat history

4.6 Functions unrelated to invoicing

Stated here because clause 7.10 admits no undocumented function, whether or not it touches a receipt.

FUNCTION	WHERE	NOTES
Analytics dashboard	Manager	Charts over sales already recorded
Assistant	Manager	Answers questions about recorded figures. Reads only; cannot alter a sale, a receipt or stock
Marketing	Manager	Campaign suggestions. No fiscal effect
QR menu studio	Manager	Publishes a guest-facing menu
Guest QR ordering	Guest browser	Creates an unconfirmed order a waiter must confirm. Cannot settle or print
Upselling report	Manager	Suggests item pairings. No fiscal effect

None of the functions in 4.6 can create, alter or delete a fiscal receipt, change a declared amount, or affect receipt numbering. They read recorded data or produce material outside the invoicing path.

5 · FISCAL OPERATION

5.1 Tax

Menu prices include VAT. Tax is extracted from the price rather than added to it, so the amount a guest is charged is the amount shown on the menu. Each item carries a tax type of A, B, C or D; unclassified items are treated as standard rated.

Each line is rounded to two decimal places and the order total is the sum of the rounded lines. Per-bracket tax is computed from the bracket total rather than by summing rounded lines, so the printed figures reconcile with one another.

5.2 Receipt numbering

Two counters are maintained per outlet: one per receipt type and one across all types, printed as `A/B RT` (clause 7.24.4). Numbers are claimed inside the settlement transaction, so a sale that fails leaves no gap. Counters never reset except on a total reset of the system, when they recommence from 1.

5.3 Receipt types

CODE	TYPE	BEHAVIOUR
NS	Normal sale	Signed. One original only
NR	Refund	Signed. References the original; permitted once per original
CS	Copy	Marked COPY with the disclaimer
TS	Training	Unsigned, marked, excluded from sales
PS	Proforma	Unsigned, marked, excluded from sales

5.4 The buyer's TIN

Where the guest is a business and asks for the receipt in the company's name, the operator records the buyer's TIN at settlement. It prints on the receipt as the client identifier and is carried to the controller as `custTin`.

It is optional and no operator is prompted for it, because most guests are individuals who have none — the field in the VSDC specification is likewise nullable. A value is accepted only as nine digits: the till will not settle on a partial entry, and the server independently declines anything that is not nine digits rather than declare a purchase against a taxpayer who may not exist. This is the buyer's TIN alone; the taxpayer's own is held against the outlet and prints on every receipt regardless.

5.5 Recovering an interrupted print

Each attempt to print a receipt is recorded with the number of lines the printer confirmed. Where printing stops part-way — paper exhausted, power lost, printer offline — the till resumes the same receipt from the line reached, or reprints it where nothing reached the paper (clause 7.28).

Whether an original ever completed is recorded rather than inferred, because clause 7.18 permits one original for each sale. Where an original did complete, any further printing of that receipt is marked as a copy. Where a sale's receipt never completed, the till does not open a further sale until the matter is resolved: a sale registered without a receipt is the condition clause 7.15 forbids.

5.6 Immutability

A settled bill cannot be cancelled, deleted or edited. The only means of reversing a sale is a refund receipt referencing the original, and an

original may be reversed once only.

5.7 Journal

Every receipt is recorded before it is printed. Should the journal be unwritable, or should any printed receipt be unrecorded, the till stops trading (clause 7.14).

5.8 Stock

A receipt is not issued for goods when the recorded stock is less than the quantity requested. Service items are exempt (clause 7.30). Items whose stock has never been counted are not treated as out of stock.

6 · CONNECTION TO THE SALES DATA CONTROLLER

This section is stated separately because it is the one part not yet verified. The message format and everything that surrounds it are built and tested. The transport and its failure behaviour require a controller instance, which has been requested from the Authority.

The till communicates with a Sales Data Controller over REST. Sales are sent to `/trnsSales/saveSales`. The response supplies the SDC identifier, the per-type and total receipt counters, the internal data and the receipt signature, which are printed in the SDC information block.

Built and verified

- The sales message itself, in the format defined by the VSDC API specification. Its figures reproduce the worked example in that specification, and the per-bracket rounding matches the printed receipt in clause 13.1.
- The journal entry written before printing, held pending until the controller answers and confirmed or failed on its response.
- Receipt numbering, claimed within the settlement so that a failed sale leaves no gap.
- The receipt layout, including the SDC information block, the internal data and signature grouped in fours, and the QR payload composed as clause 7.24.7 defines.

Awaiting a controller instance

The following are specified and cannot be completed or demonstrated without a Sales Data Controller to communicate with:

- Sending the message and receiving the response (clause 7.19, 7.20).
- Withholding the receipt until the controller has responded (clause 10, 7.8).

- Detecting the controller's availability independently and reporting its status to the user (clause 7.9).
- Ceasing operation on a communication error and resuming when it clears (clause 24.7.9).
- Transmitting journal data for sale and refund receipts (clause 6.5).

Behaviour cannot be tested against a controller that does not exist, and YoFinder Ltd will not describe as verified what it has not verified. A test instance has been requested; this section will be completed and this document revised once the integration has been demonstrated.

7 · ACCESS CONTROL

Staff sign in by PIN on shared terminals or by username and password for personal accounts. Both are stored as one-way hashes and cannot be recovered; a forgotten PIN is reset, not retrieved.

Supervisor approval, entered as a separate PIN and recorded by name, is required to cancel an order, remove a sent dish, apply a discount, move a line between bills, take cash out of the till, issue a refund, and change the TIN.

8 · DATA, RETENTION AND AUDIT

Fiscal receipts, cash movements and counter state are retained indefinitely and are never deleted by any function available to a user. Ordinary orders may be hidden from management views by date without being removed from the database.

The audit interface presents software identity and version, fiscal configuration, tax rates as programmed, record counts by receipt type, the state of each counter, and any receipt awaiting or having failed transmission.

9 · SUPPORT AND UPDATES

The software version is shown in the application, transmitted with every request to the server, and printed on the receipt. Certified installations do not update automatically; a new version is installed deliberately, so that an installation cannot move past the version RRA approved.

This documentation is revised for every change to the software, as clause 7.7 requires of the version and clause 7.10 requires of the documentation.

10 · WHAT THE SYSTEM DOES NOT DO

- It does not register a sale without printing a receipt.
- It does not print a receipt without a response from the controller.
- It does not permit a sale amount to be entered without identifying an item.

- It does not allow VAT to be disabled by any setting or user action.
- It does not allow a settled bill to be cancelled, edited or deleted.
- It does not allow an original transaction to be reversed more than once.
- It does not reset or reuse a receipt number, except on the clause 7.2 transfer-of-ownership reset, after which numbering recommences from 1 as clause 7.3 requires.
- It does not remove a fiscal record by any function other than that reset, and that reset is refused until the records have been exported to the taxpayer.
- It does not issue receipts from the tablet or the manager application.

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Revised for every change to the software. Section 6 completes on verification against a Sales Data Controller.